

Introduction to Quantum Algorithms

(Session 4 – Day 2, 4 July 2026)

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Learning Objectives

- Understand what makes a quantum algorithm faster than classical counterparts (interference and superposition).
- Study the Deutsch–Jozsa algorithm as the first example of exponential quantum speed-up for a promise problem.
- Learn the Bernstein–Vazirani algorithm and its application in extracting a hidden bitstring.
- Explore Simon’s problem and its exponential gap, which served as the key inspiration for Shor’s period-finding algorithm.
- Recognize the difference between classical and quantum query complexity.

1 Why Quantum Algorithms?

- Quantum algorithms exploit superposition and constructive/destructive interference to extract global features of a function using fewer queries than possible classically.
- Query complexity (measuring the number of calls to an oracle black box) is a standard mathematical framework used to prove separations between classical and quantum models.
- Foundational oracle algorithms (Deutsch–Jozsa, Bernstein–Vazirani, Simon) demonstrate separations that prove the computational advantage of quantum mechanics.

2 Deutsch–Jozsa Algorithm

Given a black-box function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, we are promised that f is either:

- **Constant:** The output is the same (all 0s or all 1s) for all inputs.
- **Balanced:** The output is 0 for exactly half of the inputs and 1 for the other half.

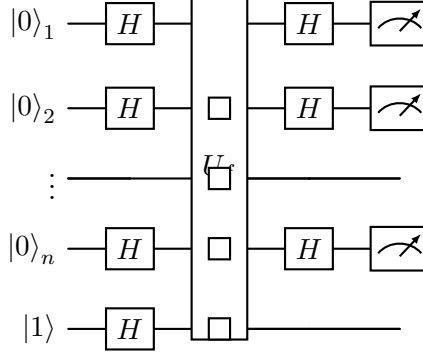
The goal is to determine if f is constant or balanced using the minimum number of queries.

2.1 Complexity Comparison

- **Classical worst-case complexity:** We must query $2^{n-1} + 1$ inputs to be 100% certain of the classification.
- **Quantum complexity:** A single query to the oracle suffices.

2.2 Quantum Circuit

The circuit diagram for the Deutsch–Jozsa algorithm is:



Here, U_f is the standard unitary oracle defined as $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$.

2.3 Step-by-Step Mathematical Analysis

1. **Initial State:** Qubits are initialized to $|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle$.
2. **Superposition Preparation:** Applying Hadamard gates to all qubits:

$$|\psi_1\rangle = (H^{\otimes n} \otimes H) |\psi_0\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \right) \otimes |-\rangle$$

3. **Oracle Query:** Applying U_f . Using the phase kickback effect, we get:

$$|\psi_2\rangle = U_f |\psi_1\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle \right) \otimes |-\rangle$$

4. **Final Hadamard Transformation:** Applying $H^{\otimes n}$ to the control register:

$$|\psi_3\rangle = (H^{\otimes n} \otimes I) |\psi_2\rangle = \left(\frac{1}{2^n} \sum_{y \in \{0,1\}^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x) + x \cdot y} |y\rangle \right) \otimes |-\rangle$$

where $x \cdot y = \sum_{i=1}^n x_i y_i \pmod{2}$ is the bitwise inner product.

5. **Measurement of Control Register:** Let us compute the amplitude of the state $|y\rangle = |0\rangle^{\otimes n}$ in the superposition:

$$C_{0^{\otimes n}} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}$$

* **If f is constant:** $(-1)^{f(x)}$ is either $+1$ for all x or -1 for all x . Thus, $C_{0^{\otimes n}} = \pm 1$, and the probability of measuring $|0\rangle^{\otimes n}$ is:

$$P(0^{\otimes n}) = 1$$

* **If f is balanced:** The sum contains exactly 2^{n-1} terms of $+1$ and 2^{n-1} terms of -1 . Thus, $C_{0^{\otimes n}} = 0$, and the probability of measuring $|0\rangle^{\otimes n}$ is:

$$P(0^{\otimes n}) = 0$$

Therefore, measuring the control register yields $|0\rangle^{\otimes n}$ if and only if f is constant. Any other outcome indicates f is balanced.

3 Bernstein–Vazirani Algorithm

The Bernstein–Vazirani algorithm finds a hidden bitstring $a \in \{0, 1\}^n$ given access to an oracle function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ defined as:

$$f(x) = a \cdot x \pmod{2}$$

3.1 Complexity Comparison

- **Classical:** Finding a requires n queries (querying $x = 100\dots 0$, $010\dots 0$, etc., to learn each bit of a individually).
- **Quantum:** A single query is sufficient.

3.2 Mathematical Analysis

The circuit is identical to the Deutsch–Jozsa circuit. 1. The state after the oracle query is:

$$|\psi_2\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{a \cdot x} |x\rangle \right) \otimes |-\rangle$$

2. Applying $H^{\otimes n}$ to the control register performs the basis rotation:

$$|\psi_3\rangle = (H^{\otimes n} \otimes I) |\psi_2\rangle = |a\rangle \otimes |-\rangle$$

3. Measuring the control register yields the exact hidden string a with 100% probability.

4 Simon's Problem

Given a black-box function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$, we are promised that there exists a hidden bitstring $s \in \{0, 1\}^n$ (the period) such that for all $x, y \in \{0, 1\}^n$:

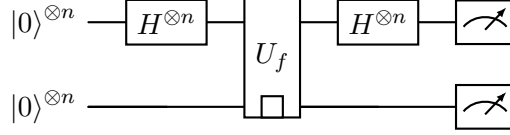
$$f(x) = f(y) \iff x \oplus y \in \{0^n, s\}$$

The goal is to determine the period s .

4.1 Complexity Comparison

- **Classical:** By the birthday paradox, finding a collision requires querying $\Omega(2^{n/2})$ inputs.
- **Quantum:** Finding s requires $\mathcal{O}(n)$ queries.

4.2 Quantum Circuit



4.3 Mathematical Analysis

1. **Prepare Superposition:** We initialize the registers to $|0\rangle^{\otimes n} |0\rangle^{\otimes n}$ and apply $H^{\otimes n}$ to the first register:

$$|\psi_1\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle \right) |0\rangle^{\otimes n}$$

2. **Oracle Query:** Applying the oracle $U_f |x\rangle |0\rangle = |x\rangle |f(x)\rangle$:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle |f(x)\rangle$$

3. **Target Measurement:** Measuring the second register yields a specific value $f(x_0)$. This collapses the first register into a superposition of the only two inputs that map to $f(x_0)$:

$$|\psi_3\rangle = \frac{1}{\sqrt{2}} (|x_0\rangle + |x_0 \oplus s\rangle)$$

4. **Final Hadamard:** Applying $H^{\otimes n}$ to the collapsed state:

$$|\psi_4\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \{0,1\}^n} \left((-1)^{x_0 \cdot y} + (-1)^{(x_0 \oplus s) \cdot y} \right) |y\rangle$$

$$|\psi_4\rangle = \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \{0,1\}^n} (-1)^{x_0 \cdot y} (1 + (-1)^{s \cdot y}) |y\rangle$$

5. **Orthogonality Condition:** Notice that the amplitude is non-zero if and only if:

$$s \cdot y \equiv 0 \pmod{2}$$

Therefore, measuring the control register yields a random bitstring y_i that is orthogonal to the hidden period s . 6. **Classical Post-Processing:** We repeat this circuit to obtain $n - 1$ linearly independent vectors $y_1, y_2, \dots, y_{n-1}$ orthogonal to s . We then solve the system of linear equations:

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n-1} \end{pmatrix} s = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \pmod{2}$$

using Gaussian elimination to determine s uniquely.

5 Discussion Problems / Exercises

1. **Bernstein–Vazirani Oracle Verification:** For a hidden string $a = 101$, construct the oracle U_f using CNOT gates and verify step-by-step how the phase kickback encodes a .

Session Takeaway: The oracle is implemented by placing CNOT gates with controls on the wires where $a_i = 1$ and target on the auxiliary qubit $|-\rangle$. The phase shift of -1 kicks back to only the control qubits that have a CNOT applied, encoding $(-1)^{a \cdot x}$.

2. **Simon’s Classical Lower Bound:** Explain why classical randomized algorithms require $\Omega(2^{n/2})$ queries to find the hidden period s in Simon’s problem.

Session Takeaway: Finding s classically requires finding a collision $f(x) = f(y)$. By the birthday paradox, the number of random queries needed to find a collision in a space of size 2^n scales as $\mathcal{O}(\sqrt{2^n}) = \mathcal{O}(2^{n/2})$.

3. **Deutsch–Jozsa Failure under Noise:** Analyze what happens to the success probability of the Deutsch–Jozsa algorithm if the oracle has a phase error that flips $(-1)^{f(x)} \rightarrow -(-1)^{f(x)}$ for a single input x_0 .

Session Takeaway: A single phase error disrupts the perfect constructive/destructive interference, causing a balanced function to yield the constant outcome $|0\rangle^{\otimes n}$ with probability $1/2^{2n-2}$, highlighting the sensitivity of quantum algorithms to phase noise.

6 Take-away Messages

1. Quantum interference allows parallel computational pathways to cancel out, leaving only the desired global answer.
2. Oracle speed-ups are formulated in the black-box query model, serving as a template for algorithms with practical applications (e.g. Shor’s period-finding).
3. The transition from Deutsch–Jozsa to Bernstein–Vazirani and Simon’s algorithm highlights a progressive structuring of functions that enables exponential computational speed-ups.

Further reading (optional)

Nielsen & Chuang, Chap. 8;

Mike & Ike, “Quantum Computation and Quantum Information”;

De Wolf, “Quantum Computing: Lecture Notes”.

A Detailed Solutions to Discussion Problems

A.1 Solution to Problem 1: Bernstein–Vazirani Oracle with $a = 101$

Let $a = 101$. Qubits 0 and 2 are the active controls. The target is qubit 3 initialized to $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$. The CNOT gates are applied from qubit 0 to 3, and qubit 2 to 3. The combined oracle is:

$$U_f = \text{CNOT}_{2,3} \text{CNOT}_{0,3}$$

Let's trace its action on a basis state $|x_2 x_1 x_0\rangle |-\rangle$: 1. The target qubit is in a superposition of $|0\rangle$ and $|1\rangle$. 2. If $x_0 = 1$, the first CNOT flips the target. If $x_2 = 1$, the second CNOT flips the target. 3. The net number of target flips is $(x_0 + x_2) \pmod{2} = a \cdot x \pmod{2}$. 4. Flipping the states of $|-\rangle$ yields:

$$X|-\rangle = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle$$

5. Thus, each flip introduces a phase of -1 :

$$U_f |x\rangle |-\rangle = (-1)^{a \cdot x} |x\rangle |-\rangle$$

This proves that CNOT gates map the classical function $a \cdot x$ directly to the phase $(-1)^{a \cdot x}$.

A.2 Solution to Problem 2: Classical Lower Bound for Simon's Problem

To find the hidden period $s \neq 0^n$ classically, we need to find two distinct inputs x and y such that $f(x) = f(y)$, which implies $x \oplus y = s$. 1. If we make k queries to the oracle, we obtain a set of k outputs $\{f(x_1), f(x_2), \dots, f(x_k)\}$. 2. The number of pairs of inputs we can form from k queries is:

$$\binom{k}{2} = \frac{k(k-1)}{2} \approx \frac{k^2}{2}$$

3. For each pair (x_i, x_j) , the probability that they collide ($f(x_i) = f(x_j)$) is $1/(2^n - 1) \approx 1/2^n$ (since each input collides with exactly one other input out of 2^n). 4. The probability of finding at least one collision after k queries is bounded by the sum of probabilities of all pairs:

$$P(\text{collision}) \leq \sum_{\text{pairs}} P(\text{pair collides}) \approx \frac{k^2}{2^{n+1}}$$

5. For the probability of success to be at least $1/2$, we require:

$$\frac{k^2}{2^{n+1}} \geq \frac{1}{2} \implies k^2 \geq 2^n \implies k \geq 2^{n/2}$$

Thus, any classical randomized algorithm requires $\Omega(2^{n/2})$ queries to find the period with high probability.

A.3 Solution to Problem 3: Deutsch–Jozsa Success under Noise

Suppose a single input x_0 has its phase flipped: $(-1)^{f(x_0)} \rightarrow -(-1)^{f(x_0)}$. The state before the final Hadamard is:

$$|\psi'_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \neq x_0} (-1)^{f(x)} |x\rangle - \frac{1}{\sqrt{2^n}} (-1)^{f(x_0)} |x_0\rangle$$

$$|\psi'_2\rangle = \left(\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \right) - \frac{2}{\sqrt{2^n}} (-1)^{f(x_0)} |x_0\rangle$$

Applying $H^{\otimes n}$ to retrieve the amplitude of $|0\rangle^{\otimes n}$:

$$C'_{0^{\otimes n}} = \left(\frac{1}{2^n} \sum_x (-1)^{f(x)} \right) - \frac{2}{2^n} (-1)^{f(x_0)}$$

1. **If f is balanced:** The first term is 0. The amplitude becomes:

$$C'_{0^{\otimes n}} = -\frac{2}{2^n} (-1)^{f(x_0)} \implies P(0^{\otimes n}) = |C'_{0^{\otimes n}}|^2 = \frac{4}{2^{2n}} = \frac{1}{2^{2n-2}}$$

Since $P(0^{\otimes n}) > 0$, there is a non-zero probability of measuring all zeros and incorrectly concluding that the function is constant. 2. **If f is constant:** The first term is ± 1 . The amplitude becomes:

$$C'_{0^{\otimes n}} = \pm 1 - \frac{2}{2^n} (\pm 1) = \pm \left(1 - \frac{2}{2^{n-1}} \right)$$

The probability of measuring all zeros is:

$$P(0^{\otimes n}) = \left(1 - \frac{1}{2^{n-1}} \right)^2 < 1$$

This degrades the success probability of identifying a constant function, demonstrating how quantum interference is highly sensitive to localized phase noise.